

Cooperative Post-Harvest Loss Tracker

Project Description — NISR 2026 Big Data Hackathon (Track 1: Agricultural Productivity)

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The Problem

After farmers harvest their crops in Rwanda, a significant portion is lost before reaching a market or a plate — to theft, pests, birds, harvesting damage, transport, poor storage, or waste during processing and packaging. This is a recognized national priority under Rwanda's NST2 development plan. Most agricultural cooperatives have no structured way to track how much they are losing, why, or how their losses compare to others nearby — making it hard to know where to focus limited resources.

Our Solution

A digital tool that helps agricultural cooperatives log their harvest and losses, see exactly how their loss rate compares to real national data for their district and crop, understand which cause is driving their losses, and receive a concrete, cause-specific recommendation for what to do about it.

- A cooperative logs harvested quantity and loss by cause (theft, pests, birds, harvesting, transport, storage, processing, packaging, sale) for a crop, district, and season.
- The tool compares their reported loss rate to real NISR survey benchmarks for their district and crop, and classifies it as Low, Medium, or High risk relative to that benchmark.
- It identifies the dominant cause behind their loss and returns a specific, practical recommendation tied to that cause — for example, improved storage practices when storage-related loss is high.
- It translates the estimated loss into Rwandan Francs using real average selling prices, so the impact is expressed in terms a cooperative leader or policymaker immediately understands.

Who It Helps

- **Cooperatives** — to understand and reduce their losses, protecting farmer income.
- **Government and development partners** — to see which districts and crops have the highest loss rates, guiding where to focus support programs.
- **Civil society organizations** — to use real, localized data when advocating for storage infrastructure or farmer training.

The Data Behind It

Built entirely on real, official NISR Seasonal Agricultural Survey microdata — both the 2024 and 2025 editions, combined for a larger and more reliable sample (over 134,000 crop-plot records, nearly 38,000 with a reported loss). The survey's questionnaire changed between the two years, so the team defined a consistent loss measure (the sum of itemized loss causes) that works across both, cleaned known data-entry issues (a district-name typo, a missing-value sentinel inflating

reported prices), and weighted all benchmarks by the survey's own sampling weights for a nationally representative result.

A Note on Methodological Honesty

The team initially tested a machine learning model to predict a precise loss percentage, validated properly by training on 2024 data and testing on unseen 2025 data. That test showed the available features could not support a reliable precise prediction. Rather than present a falsely precise number, the tool instead classifies risk into clear categories grounded in real benchmark comparisons, and bases its recommendations directly on the measured cause-of-loss breakdown. This was a deliberate choice to prioritize a trustworthy, defensible result over an impressive-looking but unreliable one.

The Technology

- Python/pandas for data cleaning and benchmark computation.
- A FastAPI backend serving benchmark lookups and cooperative risk assessments.
- A React dashboard for cooperatives to log data and view their results — risk category, benchmark comparison, cause-of-loss breakdown, and recommendation.

Why It Matters

This turns raw government statistics into something a cooperative leader can act on — a clear answer to "is this normal, why is it happening, and what should I do about it," grounded in real, verified data rather than an unreliable prediction dressed up to look sophisticated.